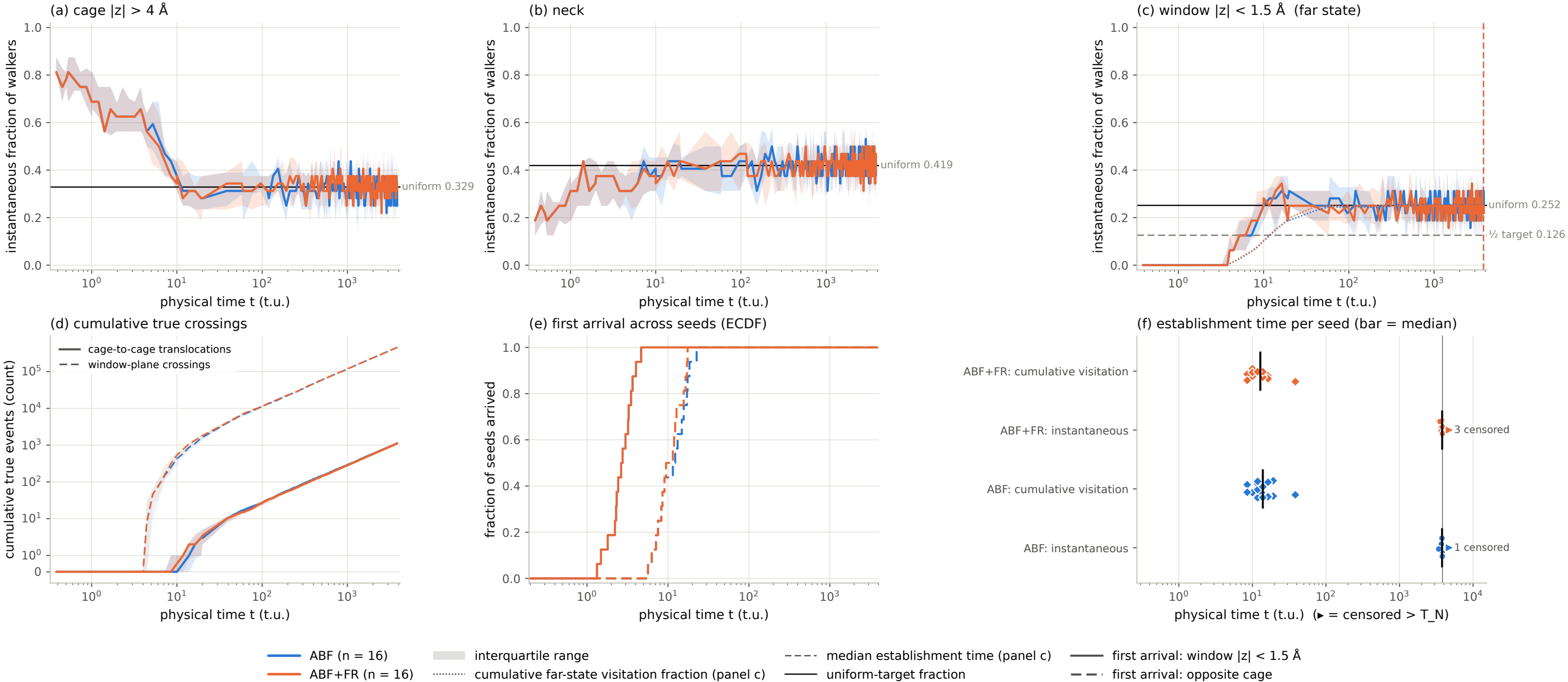


Ethane/LTA 300 K · N = 16 · T_N = 3840 t.u.

n_steps = 19,200,000 · h = 0.0002 · B = N n_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16, ABF+FR 16/16

Establishment and crossings vs physical time



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

Establishment (frozen): first save after which the instantaneous far-state fraction (window $|z| < 1.5 \text{ \AA}$) stays $\geq \frac{1}{2}$ of its uniform target 0.252; the cumulative-visitation version uses C_all (far-state bins) against $\frac{1}{2}$ of their bin share. True events never count a copy (clones inherit labels).

Null calibration at N = 16: an EXACTLY uniform population (far count $\sim \text{Bin}(N, 0.252)$ per save) fails the instantaneous criterion at a save with $p = 0.192$ and passes every save of the second half with $P = 0.000 \rightarrow$ UNRELIABLE here: the instantaneous establishment time is censored / late by chance even when established, and FR birth-death can push the count below binomial scatter (an FR advantage in it is not evidence of faster convergence); read the cumulative version and the first arrival.